

The Utility Problem

Power factor is the ratio of active to apparent electrical power in an alternating current (AC) circuit. Low power factor values at the appliance mean higher power losses on the supply line, therefore, it is a very important parameter. The closer the value is to 1, the more power is transferred and the lower the line losses are. Thus, utilities should do their best to keep power factor values in the range of 0.95 - 1 on the grid.

To achieve this, they first need to identify the areas of the grid where the power factor is poor. For this, it is necessary to have measured data from the grid and visualize it for better evaluation. However, often only raw grid measurements are taken, and the data is not processed and used in any way.

Once locations with poor power factors are found, various adjustments to the grid can be made (e.g. power factor compensators, filtering devices, etc.).

The Data Solutions and Methods

For conducting such an analysis, it is necessary to have time-series data of the power factor or time-series data of other electrical values that can be used to calculate power factor. The Awesense platform provides a simplified analysis of the power factor either across the entire grid or in selected parts of the grid. The platform's unique integration of geospatial data (GIS) with time series data (sensor data – power factor measurement) enables this type of analysis. Pre-built GIS tracing or geospatial functions speed up the creation of analyses for selected parts of the grid. Any power factor value can be displayed for a given part of the grid (e.g. average, median, maximum, etc.). These values can then be plotted on a map or table for a more comprehensive evaluation.

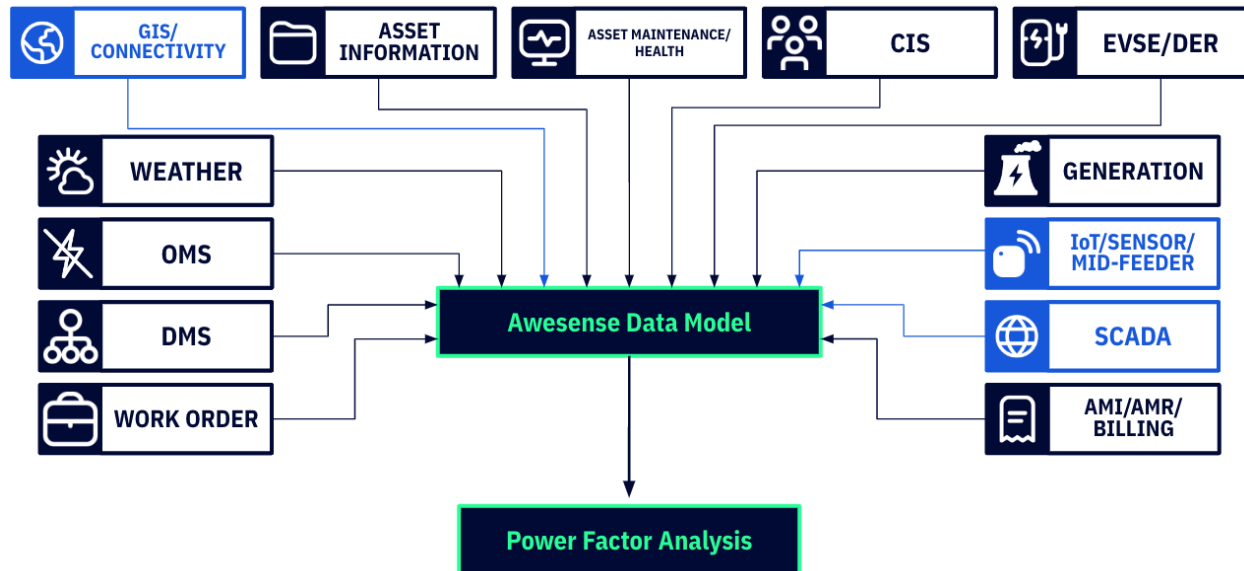
In addition to developing custom power factor analyses using the Awesense EDM interface, the Awesense TGI software includes power factor reporting. This provides information about a single location on the grid instantly and without the need to create code. In TGI, it is also possible to visualize values on a map to display multiple locations. This makes it possible to display locations in the grid with power factor values outside of the chosen values (for example, we choose to display a power factor value less than 0.5 and then these values are displayed on the map). These parameters can be flexibly changed.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

Power Factor Analysis

[UC02-01] - Awesense Energy Transition Platform Use Case



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Power Factor Analysis - Please speak to Awesense about access to the sandbox environment.